

GREATER CUMBERLAND, MD, USA

WMO: 720355

Lat: 39.617N

Lon: 78.767W

Elev: 775

StdP: 14.29

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 03732

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.1	17.5	-3.6	4.7	17.8	1.6	6.1	21.1	23.5	31.8	20.0	33.9	5.0	270	0.417

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	91.3	72.1	90.1	71.6	87.9	70.9	76.2	86.4	74.7	84.3	73.5	82.5	6.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.0	126.2	78.9	72.3	123.1	78.1	71.7	120.5	77.4	40.0	85.9	38.7	84.6	37.6	81.4	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.7	16.1	12.9	5.6	96.4	3.8	3.2	2.9	98.7	0.6	100.6	-1.6	102.4	-4.4	104.7		
			5.4	78.2	4.5	2.1	2.2	79.8	-0.4	81.0	-2.9	82.2	-6.2	83.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.5	32.2	34.7	43.3	54.1	64.0	71.0	75.2	73.5	67.4	55.9	43.8	37.1
	DBStd	17.00	8.63	9.86	9.64	8.88	7.96	5.54	4.89	4.80	6.76	8.41	7.96	7.69
	HDD50	1965	555	437	256	56	3	0	0	0	33	219	406	
	HDD65	4878	1016	847	675	341	121	14	1	2	53	307	636	865
	CDD50	3593	4	10	48	180	435	631	781	730	521	215	32	6
	CDD65	1030	0	0	1	14	89	195	317	267	124	23	0	0
	CDH74	9609	0	8	24	306	858	1776	2998	2274	1142	216	7	0
	CDH80	3520	0	0	1	95	267	597	1254	838	414	54	0	0
(14) Wind	WSAvg	4.0	4.8	5.1	5.4	5.1	3.6	3.5	3.1	2.6	2.8	3.5	4.0	4.2
(15) Precipitation	PrecAvg	36.5	2.5	2.4	3.3	3.2	3.8	3.3	3.5	3.2	3.4	2.6	2.8	2.6
	PrecMax	53.2	5.2	4.9	7.1	8.3	7.3	8.9	7.0	8.0	9.1	8.1	7.5	6.0
	PrecMin	26.6	0.4	0.1	0.7	0.7	0.4	0.8	0.6	1.2	0.1	0.2	0.1	0.6
	PrecStd	5.9	1.3	1.4	1.5	1.7	1.8	1.7	1.9	1.4	2.0	1.8	1.5	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.3	71.8	77.3	88.4	90.2	92.8	96.8	94.6	92.6	86.1	72.8	63.5
		MCWB	52.6	58.8	58.6	61.9	69.9	72.9	75.1	72.0	71.0	69.5	56.8	56.9
	2%	DB	54.7	62.9	71.9	81.9	86.5	90.0	92.7	90.6	89.7	80.5	67.6	55.4
		MCWB	47.1	51.6	57.2	61.4	68.1	72.0	73.7	72.1	70.5	66.5	55.0	49.1
	5%	DB	48.1	55.4	65.6	77.0	82.2	86.5	90.1	88.0	85.7	73.8	62.5	51.9
		MCWB	43.5	47.2	54.1	59.4	66.5	70.9	72.6	71.3	68.8	61.9	52.2	46.7
	10%	DB	44.6	49.6	60.9	71.7	79.2	82.5	87.7	85.6	81.1	70.2	57.3	47.6
		MCWB	40.1	43.0	50.0	56.2	65.2	68.6	71.9	70.8	66.7	61.3	49.3	43.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.0	59.5	62.4	67.4	74.4	76.2	78.2	77.5	75.3	72.5	60.8	57.8
		MCDB	62.7	73.0	69.2	80.3	83.3	87.7	90.7	85.6	84.0	80.2	70.8	63.2
	2%	WB	48.8	53.0	59.2	63.8	71.9	74.4	76.5	75.5	73.1	69.3	57.3	52.4
		MCDB	54.0	59.6	68.3	74.6	80.3	84.7	88.0	83.9	82.9	75.7	64.0	55.2
	5%	WB	44.1	48.1	55.9	61.3	69.9	73.0	75.0	74.0	71.6	65.9	53.6	46.6
		MCDB	47.5	56.7	64.8	71.9	76.9	82.4	85.4	82.1	80.4	73.0	59.5	49.4
	10%	WB	40.6	43.4	51.4	58.8	67.8	71.6	73.8	72.6	69.7	62.7	50.3	43.9
		MCDB	44.3	49.3	58.6	69.6	75.8	80.0	83.1	80.4	76.8	67.5	56.3	47.5
(35) Mean Daily Temperature Range	5% DB	MDBR	13.5	16.0	19.1	22.7	22.0	21.1	21.7	21.2	22.2	20.4	18.4	13.6
		MCDBR	21.8	26.8	30.2	33.8	28.3	25.3	25.6	25.5	28.5	27.7	27.2	21.1
		MCWBR	15.8	17.4	17.3	15.7	12.1	10.0	8.7	9.0	10.9	13.6	16.4	15.4
	5% WB	MCDBR	20.0	25.7	26.8	27.4	23.1	21.4	21.4	19.5	23.0	22.4	23.1	18.2
		MCWBR	15.2	17.8	16.2	14.6	10.6	9.8	8.4	8.4	9.5	11.5	15.8	14.7
(40) Clear-Sky Solar Irradiance	taub		0.308	0.322	0.351	0.393	0.433	0.493	0.518	0.516	0.424	0.366	0.323	0.306
	taud		2.453	2.384	2.384	2.297	2.202	2.063	2.010	1.995	2.239	2.406	2.495	2.486
	Ebn at Noon		273	285	287	280	270	252	245	241	259	265	266	266
	Edh at Noon		25	31	35	40	45	52	55	54	40	30	24	23
(44) All-Sky Solar Radiation	RadAvg		564	830	1134	1477	1650	1831	1800	1621	1333	958	671	485
	RadStd		54	114	73	130	156	143	107	91	167	112	69	49

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)	+0.68	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+211	+108

Nomenclature: See separate page